

DSA Master Sheet										
Sr No.	Sub No.	Topic	Subtopic	Problem Name	Link	Difficulty	Status	Notes	Revisit	
PHASE 1 : FUNDAMENTALS & LINEAR DATA STRUCTURES										
1	1	Arrays	Array Basics	Find Element at a Given Index	Link	★☆☆☆☆	Not Attempted		Not Attempted	
2	2			Min and Max in Array	Link	★☆☆☆☆	Tried (couldn't think of logic)		Revisit	
3	3			Sum of Array	Link	★☆☆☆☆	Logic Done (couldn't code)		Can Revisit	
4	4			Sum of Digits	Link	★☆☆☆☆	Code Done (but has error)		No Revisit Required	
5	5			Check If Array is Sorted	Link	★☆☆☆☆				
6	6			Alternates in Array	Link	★☆☆☆☆	Done			
7	7		Core Manipulations	Remove Duplicates from Array	Link	★☆☆☆☆				
8	8			Second Largest in Array	Link	★☆☆☆☆				
9	9			Reverse an Array	Link	★☆☆☆☆				
10	10			Missing Number	Link	★☆☆☆☆				
11	11			Segregate 0s and 1s	Link	★☆☆☆☆				
12	12		Intermediate Problems	Maximum Consecutive Ones	Link	★☆☆☆☆				
13	13			Palindromic Array	Link	★☆☆☆☆				
14	14			Move Zeros to End	Link	★☆☆☆☆				
15	15			Sort array with 0's 1's and 2's (Dutch Flag)	Link	★☆☆☆☆				
16	16			Equilibrium Point	Link	★☆☆☆☆				
17	17		Advanced Problems	Reverse Integer	Link	★☆☆☆☆				
18	18			Leaders in Array	Link	★☆☆☆☆				
19	19			Increasing Array	Link	★☆☆☆☆				
20	20			Rearrange Array Elements by Sign	Link	★☆☆☆☆				
21	21			Rotate Array by One	Link	★☆☆☆☆				
22	22	Majority Element I (Boyer - Moore)		Link	★☆☆☆☆					
23	23	Rotate Array by K steps		Link	★☆☆☆☆					
24	24	Wiggle Sort I		Link	★☆☆☆☆					
25	25	Majority Element II		Link	★☆☆☆☆					
26	26	Matrix Basics	Best Time to Buy and Sell Stock	Link	★☆☆☆☆					
27	27		Next Permutation	Link	★☆☆☆☆					
28	28		Maximum Value Of Expression	Link	★☆☆☆☆					
29	29		First Missing Positive	Link	★☆☆☆☆					
30	30	2D Arrays	Matrix Operations	Sum of elements in a matrix	Link	★☆☆☆☆				
31	31			Count Number Of Zeros	Link	★☆☆☆☆				
32	32			Count Negative Numbers in a matrix	Link	★☆☆☆☆				
33	33			Matrix Diagonal Sum	Link	★☆☆☆☆				
34	34			Addition of Two Square Matrix	Link	★☆☆☆☆				
35	35		Advanced Matrix	Multiply Matrices	Link	★☆☆☆☆				
36	36			Transpose Matrix	Link	★☆☆☆☆				
37	37			Spiral Matrix	Link	★☆☆☆☆				
38	38			ZigZag Matrix	Link	★☆☆☆☆				
39	39			Rotate Matrix (90°)	Link	★☆☆☆☆				
40	40	Basic Maths	Number Theory Basics	Determine Whether matrix can be obtained by rotation	Link	★☆☆☆☆				
41	41			Set Matrix Zeros	Link	★☆☆☆☆				
42	42	Strings	String Fundamentals	Check if a number is Armstrong	Link	★☆☆☆☆				
43	43			Print all Divisors of a Number	Link	★☆☆☆☆				
44	44			Check if a number is Prime	Link	★☆☆☆☆				
45	45			GCD / HCF of 2 numbers	Link	★☆☆☆☆				
46	46			Prime Factorisation of a Number	Link	★☆☆☆☆				
47	47			Count Primes in range L to R	Link	★☆☆☆☆				
48	48			Maximum Occuring Character	Link	★☆☆☆☆				
49	49		String Manipulation	Remove Spaces	Link	★☆☆☆☆				
50	50			Print first letter of every word in the string	Link	★☆☆☆☆				
51	51			Remove Consecutive Characters	Link	★☆☆☆☆				
52	52			Valid Palindrome	Link	★☆☆☆☆				
53	53			Valid Anagram	Link	★☆☆☆☆				
54	54			Isomorphic Strings	Link	★☆☆☆☆				
55	55			Delete Characters To Make Fancy String	Link	★☆☆☆☆				
56	56	Advanced Problems	Reverse Words in a String	Link	★☆☆☆☆					
57	57		String to integer (atoi)	Link	★☆☆☆☆					
58	58		Roman to integer	Link	★☆☆☆☆					
59	59		Rotate String	Link	★☆☆☆☆					
60	60		Longest common prefix	Link	★☆☆☆☆					
61	61		Longest palindromic substring	Link	★☆☆☆☆					
62	62		Multiple Two Strings	Link	★☆☆☆☆					
63	63	Maximum Nesting Depth of the Parentheses	Link	★☆☆☆☆						
64	64	Beauty Of All substrings	Link	★☆☆☆☆						
PHASE 2 : SEARCHING ALGORITHM										
65	1	Binary Search	BS on 1D Array Basics	Search X in sorted array	Link	★☆☆☆☆				
66	2			Lower Bound	Link	★☆☆☆☆				
67	3			Upper Bound	Link	★☆☆☆☆				
68	4			Search insert position	Link	★☆☆☆☆				
69	5			Floor in Sorted Array	Link	★☆☆☆☆				
70	6			Ceil in Sorted Array	Link	★☆☆☆☆				
71	7		Intermediate	Guess Number Higher or Lower API	Link	★☆☆☆☆				
72	8			First 1 in a Sorted Binary Array	Link	★☆☆☆☆				
73	9			Kth Missing Positive Number	Link	★☆☆☆☆				
74	10			Find minimum in Rotated Sorted Array	Link	★☆☆☆☆				
75	11			First and last occurrence	Link	★☆☆☆☆				
76	12			Search in rotated sorted array-I	Link	★☆☆☆☆				
77	13			Search in rotated sorted array-II	Link	★☆☆☆☆				
78	14		BS on 2D Arrays	Single element in a Sorted Array	Link	★☆☆☆☆				
79	15			Find kth Rotation	Link	★☆☆☆☆				
80	16			Find Peak Element	Link	★☆☆☆☆				
81	17			Count Negative Numbers in a Sorted Matrix	Link	★☆☆☆☆				
82	18			Find row with maximum 1's	Link	★☆☆☆☆				
83	19			Search a 2D Matrix I	Link	★☆☆☆☆				
84	20			Search a 2D Matrix II	Link	★☆☆☆☆				
85	21	BS on Answer	BS on Answer	Find Peak Element - II	Link	★☆☆☆☆				
86	22			Median in a row-wise sorted Matrix	Link	★☆☆☆☆				
87	23			Sign (x)	Link	★☆☆☆☆				
88	24			Valid Perfect Square	Link	★☆☆☆☆				
89	25			Find Nth root of a number	Link	★☆☆☆☆				
90	26			Koko eating bananas	Link	★☆☆☆☆				
91	27			Find the Smallest Divisor Given a Threshold	Link	★☆☆☆☆				
92	28			Minimum Speed to Arrive on Time	Link	★☆☆☆☆				
93	29			Minimum days to make M bouquets	Link	★☆☆☆☆				
94	30			Capacity to Ship Packages Within D Days	Link	★☆☆☆☆				
95	31	Sorting	Sorting Fundamentals	Book Allocation Problem	Link	★☆☆☆☆				
96	32			Split Array Largest Sum	Link	★☆☆☆☆				
97	33			Painter's Partition Problem	Link	★☆☆☆☆				
98	34			K-th element of two sorted Arrays	Link	★☆☆☆☆				
99	35			Aggressive Cows	Link	★☆☆☆☆				
100	36		Minimize Max Distance to Gas Station	Link	★☆☆☆☆					
101	37		Median of Two Sorted Arrays	Link	★☆☆☆☆					
PHASE 3 : ALGORITHMIC THINKING										
102	1	Recursion	Introduction to Recursion	Print 1 to N without using loops	Link	★☆☆☆☆				
103	2			Print N to 1 without loop	Link	★☆☆☆☆				
104	3			Sum of first N numbers	Link	★☆☆☆☆				
105	4			Factorial of a given number	Link	★☆☆☆☆				
106	5			Fibonacci Number	Link	★☆☆☆☆				
107	6		Recursion on Arrays & Math	Reverse an array	Link	★☆☆☆☆				
108	7			Power(x, n)	Link	★☆☆☆☆				
109	8			Count Good Numbers	Link	★☆☆☆☆				
110	9			Recursive Implementation of atoi()	Link	★☆☆☆☆				
111	10			Generate Binary Strings Without Consecutive 1s	Link	★☆☆☆☆				
112	1	Sorting	Sorting Fundamentals	Linear Search	Link	★☆☆☆☆				
113	2			Sort An Array (Implement Merge / Quick Sort)	Link	★☆☆☆☆				
114	3			Sort Array By Parity	Link	★☆☆☆☆				
115	4			Maximum Gap	Link	★☆☆☆☆				
116	5			Sort Elements by Decreasing Frequency	Link	★☆☆☆☆				
117	6		Advanced Sorting	Reorder Data In Log Files	Link	★☆☆☆☆				
118	7			Merge Intervals	Link	★☆☆☆☆				
119	8			Count Inversions (Merge Sort)	Link	★☆☆☆☆				
120	9			Reverse Pairs (Merge Sort)	Link	★☆☆☆☆				
PHASE 4 : OBJECT ORIENTED CONCEPTS										
121	1	OOPS	OOPS Fundamentals	Classes, Objects & Constructors	Link	★☆☆☆☆				
122	2			Inheritance & Polymorphism	Link	★☆☆☆☆				
123	3			Encapsulation & Abstraction	Link	★☆☆☆☆				
124	4		Design Patterns	Interfaces & Abstract Classes	Link	★☆☆☆☆				
125	5			Singleton Pattern	Link	★☆☆☆☆				
126	6	Factory Pattern	Link	★☆☆☆☆						
PHASE 5 : ADVANCED LINEAR DATA STRUCTURES										
127	1	Linkedlist	Foundation	Array to Linked List	Link	★☆☆☆☆				
128	2			Count Nodes / Find Length of Linked List	Link	★☆☆☆☆				
129	3			Search in a Linked List	Link	★☆☆☆☆				
130	4			Insert Node at Head / Tail	Link	★☆☆☆☆				
131	5			Insert in Middle of Linked List	Link	★☆☆☆☆				
132	6		Insertion and Deletion	Remove Linked List Elements (By Value)	Link	★☆☆☆☆				
133	7			Remove Duplicates from Sorted List	Link	★☆☆☆☆				
134	8			Delete Node in a Linked List (Given only node ref)	Link	★☆☆☆☆				
135	9			Remove Duplicates from Unsorted List	Link	★☆☆☆☆				
136	10			Delete the Middle Node of a Linked List	Link	★☆☆☆☆				
137	11		Fast and Slow Pointer	Middle of the Linked List	Link	★☆☆☆☆				
138	12			Linked List Cycle	Link	★☆☆☆☆				
139	13			Intersection of Two Linked Lists	Link	★☆☆☆☆				
140	14			Length of Loop in Linked List	Link	★☆☆☆☆				
141	15			Linked List Cycle II (Find Starting Point)	Link	★☆☆☆☆				
142	16		Reversals and Rotations	Remove Loop in Linked List	Link	★☆☆☆☆				
143	17			Remove Nth Node From End of List	Link	★☆☆☆☆				
144	18			Reverse a Linked List (Iterative & Recursive)	Link	★☆☆☆☆				
145	19			Palindrome Linked List	Link	★☆☆☆☆				
146	20		Linkedlist	Reversals and Rotations	Swap Nodes in Pairs	Link	★☆☆☆☆			
147	21				Odd Even Linked List (By Index)	Link	★☆☆☆☆			
148	22				Reorder List	Link	★☆☆☆☆			
149	23				Rotate List	Link	★☆☆☆☆			
150	24		Math in Linked List	Math in Linked List	Add One to a Number Represented by LL	Link	★☆☆☆☆			
151	25				Add Two Numbers (Reverse Order)	Link	★☆☆☆☆			
152	26				Add Two Numbers II (Forward Order)	Link	★☆☆☆☆			
153	27				Merge Two Sorted Lists	Link	★☆☆☆☆			
154	28	Sort a Linked List of 0s, 1s, and 2s	Link	★☆☆☆☆						

155	29		Sorting, Merging & Segregating	Segregate Even and Odd Nodes (By Value)	Link	★★★★☆				
156	30			Partition List	Link	★★★★☆				
157	31			Insertion Sort List	Link	★★★★☆				
158	32			Sort List (Merge Sort on LL)	Link	★★★★☆				
159	33			Array to Doubly Linked List	Link	★★★★☆				
160	34			Insert/Delete in a Doubly Linked List	Link	★★★★☆				
161	35			Reverse a Doubly Linked List	Link	★★★★☆				
162	36			Remove Duplicates from Sorted DLL	Link	★★★★☆				
163	37			Insert in Sorted way in a Sorted DLL	Link	★★★★☆				
164	38			Delete all occurrences of a key in DLL	Link	★★★★☆				
165	39			Find Pairs with Given Sum in a DLL	Link	★★★★☆				
166	40			Copy List with Random Pointer (Clone LL)	Link	★★★★☆				
167	41			Flattening a Linked List	Link	★★★★☆				
168	42			Reverse Linked List in Groups of Size K	Link	★★★★☆				
169	43			Reverse Alternate K Nodes	Link	★★★★☆				
170	1			Implement Stack using Arrays	Link	★★★★☆				
171	2			Core Implementation	Link	★★★★☆				
172	3			Implement Stack using Linked List	Link	★★★★☆				
173	4			Valid Parentheses	Link	★★★★☆				
174	5			Remove Outermost Parentheses	Link	★★★★☆				
175	6			Remove All Adjacent Duplicates In String	Link	★★★★☆				
176	7			Minimum Remove to Make Valid Parentheses	Link	★★★★☆				
177	8			Evaluate Reverse Polish Notation	Link	★★★★☆				
178	9			Recursion Based Stack Problems	Link	★★★★☆				
179	10			Reverse a Stack (Using Recursion)	Link	★★★★☆				
180	11			Postfix to Prefix Conversion	Link	★★★★☆				
181	12			Expression Conversions	Link	★★★★☆				
182	13			Prefix to Postfix Conversion	Link	★★★★☆				
183	14			Infix to Prefix Conversion	Link	★★★★☆				
184	15			Next Greater Element I	Link	★★★★☆				
185	16			Next Smaller Element / Help Classmates	Link	★★★★☆				
186	17			Next Greater Element II (Circular Array)	Link	★★★★☆				
187	18			Stock Span Problem	Link	★★★★☆				
188	19			Number of Greater Elements to the Right	Link	★★★★☆				
189	20			Adelson-Collison	Link	★★★★☆				
190	21			Remove K Digits	Link	★★★★☆				
191	22			Remove Duplicate Letters	Link	★★★★☆				
192	23			Largest Rectangle in Histogram	Link	★★★★☆				
193	24			Sum of Subarray Minimums	Link	★★★★☆				
194	25			Sum of Subarray Ranges	Link	★★★★☆				
195	26			Maximal Rectangle (2D Grid)	Link	★★★★☆				
196	27			Min Stack (Design a stack with O(1) peek/pop)	Link	★★★★☆				
197	28			Classic Design Problems	Link	★★★★☆				
198	1			The Celebrity Problem (Elimination via Stack)	Link	★★★★☆				
199	2			Implement Queue using Arrays	Link	★★★★☆				
200	3			Implement Queue using Linked List	Link	★★★★☆				
201	4			Implement Queue using Stacks	Link	★★★★☆				
202	5			Implement Stack using Queues	Link	★★★★☆				
203	6			Reverse First K elements of Queue	Link	★★★★☆				
204	7			Number of Students Unable to Eat Lunch	Link	★★★★☆				
205	8			First non-repeating character in a stream	Link	★★★★☆				
206	9			Sliding Window Maximum	Link	★★★★☆				
207	10			Longest Continuous Subarray With Abs Diff ≤ Limit	Link	★★★★☆				
208	11			Constrained Subsequence Sum	Link	★★★★☆				
PHASE 6 - HIERARCHIAL DATA STRUCTURES										
209	2			Preorder Traversal (Recursive & Iterative)	Link	★★★★☆				
210	3			Inorder Traversal (Recursive & Iterative)	Link	★★★★☆				
211	4			Postorder Traversal (Recursive & Iterative)	Link	★★★★☆				
212	5			Binary Tree Level Order Traversal	Link	★★★★☆				
213	6			Binary Tree Zigzag Level Order Traversal	Link	★★★★☆				
214	7			Left / Right View of Binary Tree	Link	★★★★☆				
215	8			Top / Bottom View of Binary Tree	Link	★★★★☆				
216	9			Boundary Traversal of Binary Tree	Link	★★★★☆				
217	10			Diagonal Traversal of Binary Tree	Link	★★★★☆				
218	11			Vertical Order Traversal of a Binary Tree	Link	★★★★☆				
219	12			Maximum Depth of Binary Tree	Link	★★★★☆				
220	13			Same Tree	Link	★★★★☆				
221	14			Invert Binary Tree	Link	★★★★☆				
222	15			Symmetric Tree	Link	★★★★☆				
223	16			Maximum Width of Binary Tree	Link	★★★★☆				
224	17			Count Complete Tree Nodes	Link	★★★★☆				
225	18			Diameter of Binary Tree	Link	★★★★☆				
226	19			Balanced Binary Tree	Link	★★★★☆				
227	20			Path Sum I	Link	★★★★☆				
228	21			Path Sum II	Link	★★★★☆				
229	22			Sum Root to Leaf Numbers	Link	★★★★☆				
230	23			Maximum Difference Between Node and Ancestor	Link	★★★★☆				
231	24			Lowest Common Ancestor of a Binary Tree	Link	★★★★☆				
232	25			Binary Tree Maximum Path Sum	Link	★★★★☆				
233	26			Path Sum III	Link	★★★★☆				
234	27			Minimum time taken to burn the BT	Link	★★★★☆				
235	28			All Nodes Distance K in Binary Tree	Link	★★★★☆				
236	29			Step-By-Step Directions From a Binary Tree Node to Another	Link	★★★★☆				
237	30			Merge Two Binary Trees	Link	★★★★☆				
238	31			Children Sum Property in a Binary Tree	Link	★★★★☆				
239	32			Construct Binary Tree from Preorder and Inorder Traversal	Link	★★★★☆				
240	33			Construct Binary Tree from Inorder and Postorder Traversal	Link	★★★★☆				
241	34			Flatten Binary Tree to Linked List	Link	★★★★☆				
242	1			Serialize and Deserialize Binary Tree	Link	★★★★☆				
243	2			Search in a Binary Search Tree	Link	★★★★☆				
244	3			Find Minimum/Maximum in BST	Link	★★★★☆				
245	4			Insert into a Binary Search Tree	Link	★★★★☆				
246	5			Floor and Ceil in a BST	Link	★★★★☆				
247	6			Inorder Successor in BST	Link	★★★★☆				
248	7			Lowest Common Ancestor of a Binary Search Tree	Link	★★★★☆				
249	8			Validate Binary Search Tree	Link	★★★★☆				
250	9			kth Smallest Element in a BST	Link	★★★★☆				
251	10			Delete Node in a BST	Link	★★★★☆				
252	11			Two Sum IV - Input is a BST	Link	★★★★☆				
253	12			Convert Sorted Array to Binary Search Tree	Link	★★★★☆				
254	13			Construct BST from Preorder Traversal	Link	★★★★☆				
255	14			Balance a Binary Search Tree	Link	★★★★☆				
256	15			Merge Two Balanced Binary Search Trees	Link	★★★★☆				
257	16			Recover Binary Search Tree	Link	★★★★☆				
258	1			Largest BST in a Binary Tree	Link	★★★★☆				
259	2			Implement Trie (Prefix Tree)	Link	★★★★☆				
260	3			Design Add and Search Words Data Structure	Link	★★★★☆				
261	4			Longest Word with All Prefixes	Link	★★★★☆				
262	5			Number of Distinct Substrings in a String	Link	★★★★☆				
263	6			Word Search II	Link	★★★★☆				
264	7			Maximum XOR of Two Numbers in an Array	Link	★★★★☆				
265	8			Maximum XOR With an Element From Array	Link	★★★★☆				
PHASE 7 - CORE DATA STRUCTURES AND TECHNIQUES										
266	1			Contains Duplicate	Link	★★★★☆				
267	2			Valid Anagram (Hashmap Approach)	Link	★★★★☆				
268	3			Unique Number of Occurrences	Link	★★★★☆				
269	4			Find distinct elements / Find the Frequency	Link	★★★★☆				
270	5			Two Sum	Link	★★★★☆				
271	6			Intersection of Two Arrays	Link	★★★★☆				
272	7			Count Number of Pairs With Absolute Difference K	Link	★★★★☆				
273	8			Design HashMap	Link	★★★★☆				
274	9			Group Anagrams	Link	★★★★☆				
275	10			Longest Consecutive Sequence	Link	★★★★☆				
276	11			Subarray Sum Equals K	Link	★★★★☆				
277	12			Contiguous Array / Largest subarray with 0 sum	Link	★★★★☆				
278	13			Count subarrays with given XOR	Link	★★★★☆				
279	14			Subarray Sums Divisible by K	Link	★★★★☆				
280	15			Continuous Subarray Sum	Link	★★★★☆				
281	16			Maximum Size Subarray Sum Equals k	Link	★★★★☆				
282	17			Top K Frequent Elements	Link	★★★★☆				
283	18			LRU Cache (Least Recently Used)	Link	★★★★☆				
284	19			LFU Cache (Least Frequently Used)	Link	★★★★☆				
285	1			Check if an array represents a min heap	Link	★★★★☆				
286	2			Convert Min Heap to Max Heap	Link	★★★★☆				
287	3			Implement Min/Max Heap	Link	★★★★☆				
288	4			Least Stone Weight	Link	★★★★☆				
289	5			kth Largest Element in an Array	Link	★★★★☆				
290	6			k Closest Points to Origin	Link	★★★★☆				
291	7			Sort a K Sorted Array / Nearly Sorted Algorithm	Link	★★★★☆				
292	8			Top K Frequent Elements (Heap Approach)	Link	★★★★☆				
293	9			Minimum Cost of Ropes	Link	★★★★☆				
294	10			Merge K Sorted Lists	Link	★★★★☆				
295	11			kth Smallest Element in a Sorted Matrix	Link	★★★★☆				
296	12			Task Scheduler	Link	★★★★☆				
297	13			Reorganize String	Link	★★★★☆				
298	14			Minimum Number of Refueling Stops	Link	★★★★☆				
299	15			Find Median from Data Stream	Link	★★★★☆				
300	1			Running Sum of 1D Array	Link	★★★★☆				
301	2			Find the Highest Altitude	Link	★★★★☆				
302	3			Find Pivot Index	Link	★★★★☆				
303	4			Range Sum Query - Immutable	Link	★★★★☆				
304	5			Product of Array Except Self	Link	★★★★☆				
305	6			Corporate Flight Bookings	Link	★★★★☆				
306	7			Trapping Rain Water (Prefix Max / Min)	Link	★★★★☆				
307	8			Range Sum Query 2D - Immutable	Link	★★★★☆				
308	9			Matrix Block Sum	Link	★★★★☆				
309	10			Maximum Sum of a 2D Subgrid of size K x K	Link	★★★★☆				
310	1			Move Zeros (Two Pointer Intuition)	Link	★★★★☆				
311	2			Valid Palindrome (Two Pointers)	Link	★★★★☆				
312	3			Two Sum II - Input Array is Sorted	Link	★★★★☆				
313	4			Sort Colors (Dutch National Flag)	Link	★★★★☆				
314	5			Container With Most Water	Link	★★★★☆				
315	6			Sum	Link	★★★★☆				
316	7			Maximum Average Subarray I	Link	★★★★☆				

316	8			Fixed Size Sliding Window	Max Sum Subarray of size K	Link	★★★★☆				
317	9				Number of Sub-arrays of Size K and Average Greater than or Equal to Average of All	Link	★★★★☆				
318	10				Minimum Consecutive Cards to Pick Up	Link	★★★★☆				
319	11				Maximum Points You Can Obtain from Cards	Link	★★★★☆				
320	12			Two Pointers & Sliding Window	Find All Anagrams in a String	Link	★★★★☆				
321	13				Permutation in String	Link	★★★★☆				
322	14				Minimum Size Subarray Sum	Link	★★★★☆				
323	15			Variable Size Sliding Window	Longest Substring Without Repeating Characters	Link	★★★★☆				
324	16				Longest Repeating Character Replacement	Link	★★★★☆				
325	17				Max Consecutive Ones III	Link	★★★★☆				
326	18				Fruit Into Baskets	Link	★★★★☆				
327	19				Maximum Erasure Value	Link	★★★★☆				
328	20				Number of Substrings Containing All Three Characters	Link	★★★★☆				
329	21				Binary Subarrays With Sum	Link	★★★★☆				
330	22				Count Number of Nice Subarrays	Link	★★★★☆				
331	23			Advanced Sliding Window / Counting / Almost K	Subarrays with K Different Integers	Link	★★★★☆				
332	24				Minimum Window Subsequence	Link	★★★★☆				
333	25				Minimum Window Substring	Link	★★★★☆				
334	1				Kth Bit is Set or Not	Link	★★★★☆				
335	2				Check if a Number is Odd or Not	Link	★★★★☆				
336	3			Bit Basics & Properties	Check if Number Power of 2 or Not	Link	★★★★☆				
337	4				Number Of Even and Odd Bits	Link	★★★★☆				
338	5				Minimum Bit Flips To Convert Number	Link	★★★★☆				
339	6				Swap Two Numbers (XOR Trick)	Link	★★★★☆				
340	7				Single Number I	Link	★★★★☆				
341	8			Bit Manipulation	Is Binary Number Multiple of 3	Link	★★★★☆				
342	9			XOR Tricks	Find the repeating and missing number	Link	★★★★☆				
343	10				Single Number II	Link	★★★★☆				
344	11				Single Number III	Link	★★★★☆				
345	12				Count Set Bits From 1 to N	Link	★★★★☆				
346	13			Bit Counting & Advanced	Break Numbers	Link	★★★★☆				
347	14				Minimum XOR Pair	Link	★★★★☆				
348	15				Divide Two Integers (Bit Shifting)	Link	★★★★☆				
PHASE 8 : STANDARD ALGORITHMS											
349	1				Assign Cookies	Link	★★★★☆				
350	2				Lemonade Change	Link	★★★★☆				
351	3			Basics & Array Manipulation	Maximize Sum of Array After K Negations	Link	★★★★☆				
352	4				Shortest Job First	Link	★★★★☆				
353	5				Activity Selection / N Meetings in One Room	Link	★★★★☆				
354	6				Merge Intervals (Greedy View)	Link	★★★★☆				
355	7				Insert Interval	Link	★★★★☆				
356	8				Non-overlapping Intervals	Link	★★★★☆				
357	9			Intervals & Scheduling	Minimum Number of Arrows to Burst Balloons	Link	★★★★☆				
358	10				Job Sequencing Problem	Link	★★★★☆				
359	11				Minimum Platforms required for a railway	Link	★★★★☆				
360	12				Maximum Meetings in One Room	Link	★★★★☆				
361	13				Task Scheduler (Greedy Approach)	Link	★★★★☆				
362	14				Jump Game	Link	★★★★☆				
363	15			Array & Jump Greedy	Valid Parenthesis String	Link	★★★★☆				
364	16				Gas Station	Link	★★★★☆				
365	17				Jump Game II	Link	★★★★☆				
366	18				Minimize the Heights II	Link	★★★★☆				
367	19			Advanced Greedy	Candy	Link	★★★★☆				
368	20				Huffman Decoding / Coding	Link	★★★★☆				
369	21				Minimum Number of Taps to Open to Water a Garden	Link	★★★★☆				
370	22				Course Schedule III	Link	★★★★☆				
371	1				BFS of Graph	Link	★★★★☆				
372	2			Graphs Fundamentals & Traversals	DFS of Graph	Link	★★★★☆				
373	3				Find if Path Exists in Graph	Link	★★★★☆				
374	4				Number of Provinces	Link	★★★★☆				
375	5				Clone Graph	Link	★★★★☆				
376	6				Is Graph Bipartite?	Link	★★★★☆				
377	7				Flood Fill	Link	★★★★☆				
378	8				Max Area of Island	Link	★★★★☆				
379	9				Number of Islands	Link	★★★★☆				
380	10				Rotting Oranges	Link	★★★★☆				
381	11			Matrix / Grid BFS Problems	01 Matrix / Distance of nearest cell having 1	Link	★★★★☆				
382	12				Surrounded Regions	Link	★★★★☆				
383	13				Number of Enclaves	Link	★★★★☆				
384	14				Shortest Path in Binary Matrix	Link	★★★★☆				
385	15				Swim in Rising Water	Link	★★★★☆				
386	16				Making A Large Island	Link	★★★★☆				
387	17				Cycle Detection in Undirected Graph	Link	★★★★☆				
388	18				Detect Cycle in a Directed Graph	Link	★★★★☆				
389	19				Find Eventual Safe States	Link	★★★★☆				
390	20				Topological Sort	Link	★★★★☆				
391	21				Course Schedule	Link	★★★★☆				
392	22			Topological Sorting (Kahn's Algorithm)	Course Schedule II	Link	★★★★☆				
393	23				Alien Dictionary	Link	★★★★☆				
394	24				Parallel Courses III	Link	★★★★☆				
395	25				Shortest Path in Undirected Graph (Unit Weights)	Link	★★★★☆				
396	26				Shortest Path in DAG	Link	★★★★☆				
397	27				Dijkstra's Algorithm	Link	★★★★☆				
398	28				Network Delay Time	Link	★★★★☆				
399	29				Path With Minimum Effort	Link	★★★★☆				
400	30			Shortest Path Algorithms (Dijkstra, Bellman-Ford, Floyd-Warshall)	Cheapest Flights Within K Stops	Link	★★★★☆				
401	31				Path with Maximum Probability	Link	★★★★☆				
402	32				Bellman-Ford Algorithm	Link	★★★★☆				
403	33				Floyd-Warshall Algorithm	Link	★★★★☆				
404	34				Find the City With the Smallest Number of Neighbors	Link	★★★★☆				
405	35				Minimum Multiplications to Reach End	Link	★★★★☆				
406	36				Number of Ways to Arrive at Destination	Link	★★★★☆				
407	37			Minimum Spanning Tree	Minimum Spanning Tree	Link	★★★★☆				
408	38				Prim's Algorithm	Link	★★★★☆				
409	39				Min Cost to Connect All Points (Kruskal)	Link	★★★★☆				
410	40				Disjoint Set	Link	★★★★☆				
411	41				Number of Operations to Make Network Connected	Link	★★★★☆				
412	42				Accidents Merge	Link	★★★★☆				
413	43			Disjoint Set Union (DSU) / Union Find	Most Stones Removed with Same Row or Column	Link	★★★★☆				
414	44				Redundant Connection	Link	★★★★☆				
415	45				Evaluate Division	Link	★★★★☆				
416	46				Number of Islands II	Link	★★★★☆				
417	47				Kosaraju's Algorithm	Link	★★★★☆				
418	48			Bridges, Articulation Points & Strongly Connected Components	Tarjan's Algorithm / Bridges in Graph	Link	★★★★☆				
419	49				Articulation Point in Graph	Link	★★★★☆				
420	50				Critical Connections in a Network	Link	★★★★☆				
421	51				Snake and Ladder Problem	Link	★★★★☆				
422	52			Advanced Graphs Problems	Word Ladder	Link	★★★★☆				
423	53				Reorder Routes to Make All Paths Lead to the City Zero	Link	★★★★☆				
424	54				Word Ladder II	Link	★★★★☆				
425	55				Escape the Spreading Fire	Link	★★★★☆				
PHASE 9 : EXHAUSTIVE SEARCH & OPTIMISATION											
426	1				Check if subsequence with sum K Exists	Link	★★★★☆				
427	2			Subsequence & Subset Fundamentals	Count all subsequences with sum K	Link	★★★★☆				
428	3				Subsets / Power Set	Link	★★★★☆				
429	4				Subsets II (With Duplicates)	Link	★★★★☆				
430	5				Subset Sums	Link	★★★★☆				
431	6			Combinations	Combination Sum I	Link	★★★★☆				
432	7				Combination Sum II	Link	★★★★☆				
433	8				Combination Sum III	Link	★★★★☆				
434	9				N Digit numbers with digits in increasing order	Link	★★★★☆				
435	10				Permutations II (With Duplicates)	Link	★★★★☆				
436	11			Backtracking	Letter Combinations of a Phone Number	Link	★★★★☆				
437	12				Generate Parentheses	Link	★★★★☆				
438	13				Palindromic Partitioning	Link	★★★★☆				
439	14			String Backtracking	Word Break	Link	★★★★☆				
440	15				Remove Invalid Parentheses	Link	★★★★☆				
441	16				Expression Add Operators	Link	★★★★☆				
442	17				Rat in a Maze	Link	★★★★☆				
443	18				Word Search	Link	★★★★☆				
444	19				N-Coloring Problem	Link	★★★★☆				
445	20			Grid & Maze Backtracking	N-Queens	Link	★★★★☆				
446	21				N-Queens II	Link	★★★★☆				
447	22				Sudoku Solver	Link	★★★★☆				
448	1				Fibonacci Number	Link	★★★★☆				
449	2				Climbing Stairs	Link	★★★★☆				
450	3			1D DP (Intro to State & Transitions)	Min Cost Climbing Stairs	Link	★★★★☆				
451	4				Frog Jump	Link	★★★★☆				
452	5				Frog Jump with K Distances	Link	★★★★☆				
453	6				House Robber	Link	★★★★☆				
454	7				House Robber II	Link	★★★★☆				
455	8				Decode Ways	Link	★★★★☆				
456	9				Unique Paths	Link	★★★★☆				
457	10				Unique Paths II	Link	★★★★☆				
458	11			2D Grids & Paths	Minimum Path Sum	Link	★★★★☆				
459	12				Triangle	Link	★★★★☆				
460	13				Minimum Falling Path Sum	Link	★★★★☆				
461	14				Dungeon Game	Link	★★★★☆				
462	15				Cherry Pickup II / Ninja and his Friends	Link	★★★★☆				
463	16				Subset Sum Problem	Link	★★★★☆				
464	17				Partition Equal Subset Sum	Link	★★★★☆				
465	18				Count Subsets with Sum K	Link	★★★★☆				
466	19			0/1 Knapsack & Subsets	Target Sum	Link	★★★★☆				
467	20				Count Partitions with Given Difference	Link	★★★★☆				
468	21				0/1 Knapsack Problem	Link	★★★★☆				
469	22				Partition Array Into Two Arrays to Minimize Sum Difference	Link	★★★★☆				
470	23				Last Stone Weight II	Link	★★★★☆				
471	24				Root Cutting Problem	Link	★★★★☆				
472	25				Coin Change	Link	★★★★☆				

476	29		Print Longest Common Subsequence	Link	★★★★☆							
477	30		Longest Common Substring	Link	★★★★☆							
478	31		Longest Palindromic Subsequence	Link	★★★★☆							
479	32		Minimum Insertions to Make String Palindrome	Link	★★★★☆							
480	33		Delete Operation for Two Strings	Link	★★★★☆							
481	34		Shortest Common Supersequence	Link	★★★★☆							
482	35		Distinct Subsequences	Link	★★★★☆							
483	36		Edit Distance	Link	★★★★☆							
484	37		Wildcard Matching	Link	★★★★☆							
485	38		Regular Expression Matching	Link	★★★★★							
486	39		Best Time to Buy and Sell Stock	Link	★★☆☆☆							
487	40		Best Time to Buy and Sell Stock II	Link	★★☆☆☆							
488	41		Best Time to Buy and Sell Stock with Cooldown	Link	★★☆☆☆							
489	42		Best Time to Buy and Sell Stock with Transaction Fee	Link	★★★★☆							
490	43		Best Time to Buy and Sell Stock III	Link	★★★★☆							
491	44		Best Time to Buy and Sell Stock IV	Link	★★★★☆							
492	45		Longest Increasing Subsequence	Link	★★★★☆							
493	46		Print Longest Increasing Subsequence	Link	★★★★☆							
494	47		Largest Divisible Subset	Link	★★★★☆							
495	48		Longest String Chain	Link	★★★★☆							
496	49		Longest Bitonic Subsequence	Link	★★★★☆							
497	50		Number of Longest Increasing Subsequences	Link	★★★★☆							
498	51		Russian Doll Envelopes	Link	★★★★★							
499	52		Matrix Chain Multiplication	Link	★★★★☆							
500	53		Minimum Cost to Cut a Stick	Link	★★★★☆							
501	54		Burst Balloons	Link	★★★★★							
502	55		Palindrome Partitioning II	Link	★★★★★							

DSA ROADMAP — LEGEND & INSTRUCTIONS

DIFFICULTY SCALE

★★★★★ (1)	Very Easy — Absolute beginner-friendly, core syntax level
★★★★★ (2)	Easy — Foundation problems, minor logical thinking required
★★★★★ (3)	Medium — Standard interview-level, requires pattern knowledge
★★★★★ (4)	Hard — Multi-concept, optimised approach needed
★★★★★ (5)	Very Hard — FAANG-level, advanced techniques required

COURSE PHASES

Phase 1	Fundamentals & Linear DS - Arrays, 2D Arrays, Math, Strings
Phase 2	Searching - Binary Search
Phase 3	Algorithmic Thinking - Recursion, Sorting
Phase 4	OOP Concepts - OOPS
Phase 5	Advanced Linear DS - Linked List, Stacks, Queues
Phase 6	Hierarchical DS - Binary Trees, BST, Tries
Phase 7	Core DS & Techniques - Hashmaps, Heaps, Prefix Sum, Sliding Window, Bit Manipulation
Phase 8	Standard Algorithms - Greedy, Graphs
Phase 9	Exhaustive Search & Optimisation - Backtracking, Dynamic Programming

PLACEMENT RULES APPLIED

Dependency Rule	Multi-concept problems are placed under the topic that appears LATER in the course
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DSA ROADMAP — TOPIC SUMMARY DASHBOARD

#	Topic	★1	★2	★3	★4	★5	Total
1	Arrays	6	9	7	6	1	29
2	2D Arrays	0	4	3	4	1	12
3	Basic Math	4	1	1	0	0	6
4	Strings	2	5	6	4	0	17
5	Binary Search	0	15	14	5	3	37
6	Recursion	5	3	2	0	0	10
7	Sorting	1	2	3	1	2	9
8	OOPS	1	2	2	1	0	6
9	Linked List	4	13	16	7	3	43
10	Stacks	2	8	12	5	1	28
11	Queues	2	4	1	2	1	10
12	Binary Trees	0	0	9	15	10	34
13	Binary Search Trees	0	2	3	10	8	23
14	Hashmaps	4	3	10	1	1	19
15	Heaps	0	2	3	5	5	15
16	Prefix Sum	1	3	3	3	0	10
17	Sliding Window	2	6	14	2	1	25
18	Bit Manipulation	2	6	1	5	1	15
19	Greedy	1	4	12	4	1	22
20	Graphs	2	10	29	12	2	55
21	Backtracking	0	2	10	5	5	22
22	Dynamic Programming	3	8	26	14	4	55
TOTAL		42	112	187	111	50	502